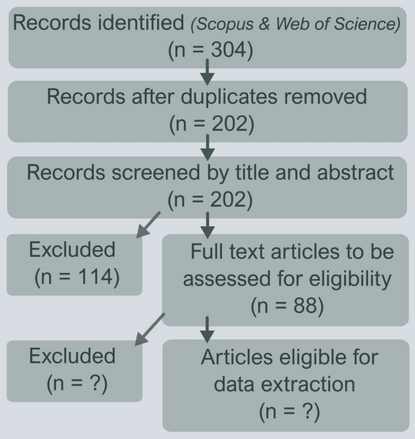
Assessing the impacts of invasive black bass (*Micropterus* spp.) on food webs in Cederberg rivers

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1 Systematic review



Stable isotope analysis ($\delta^{13}C$ and $\delta^{15}N$) quantifies the sources of energy in freshwater food webs and the trophic positions of consumers, making it a valuable tool for assessing the effects of invasive predators. However, the extent and distribution of its application in African freshwater ecosystems has not been systematically assessed.

A systematic review and meta-analysis is underway to address this gap. Searches of Scopus and Web of Science yielded 202 unique records, which are being screened in accordance with PRISMA guidelines.[5] Data extracted from each included study comprise location, ecosystem type, taxa sampled and isotope metrics.

Research Questions:

Primary (meta-analysis): In African freshwater ecosystems, how do disturbances such as invasive species, land-use change, flow alteration and drought alter food web structure, measured by community-wide isotopic metrics and isotopic niche size, relative to undisturbed or less-disturbed reference systems?

Secondary (systematic map): How are stable isotope food web studies distributed across African regions, ecosystem types and taxa, and where are the main evidence gaps?

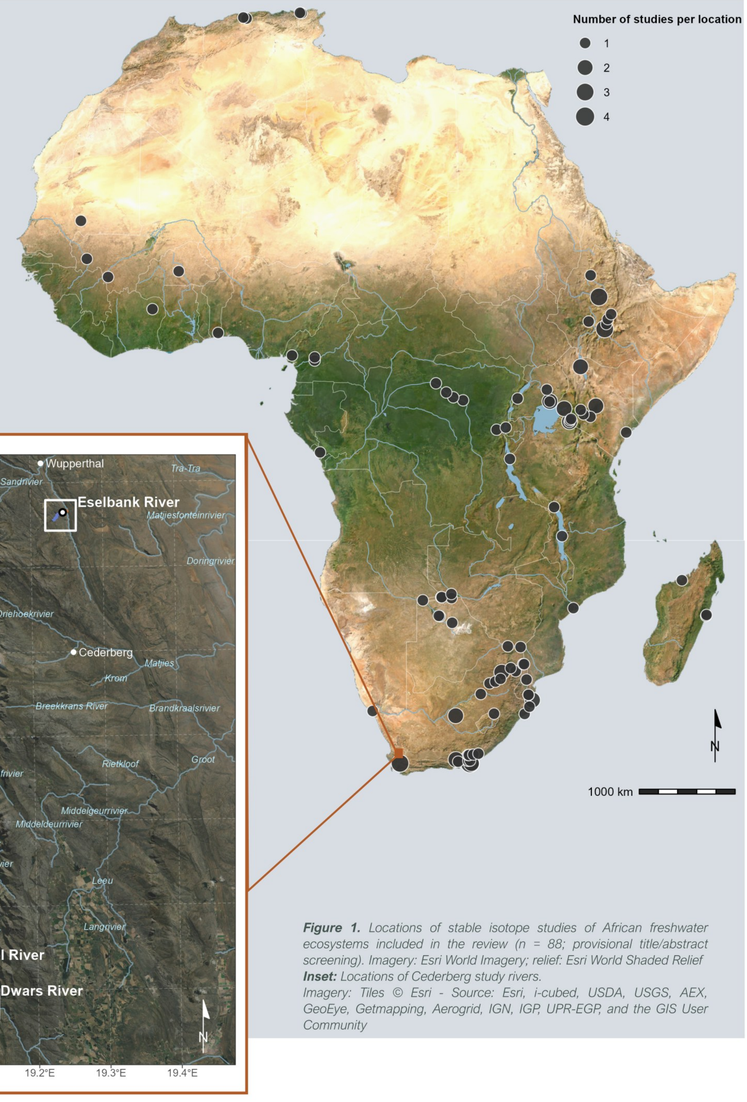


Figure 1. *Locations of stable isotope studies of African freshwater ecosystems included in the review (n = 88; provisional title/abstract screening). Imagery: Esri World Imagery; relief: Esri World Shaded Relief*
Inset: *Locations of Cederberg study rivers.*
Imagery: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

Invasive fish are among the greatest threats to the endemic freshwater fauna of South Africa's Cape Fold Ecoregion[1,3,4], yet their effects on invertebrates and food web structure remain poorly understood. This study aims to address these gaps through the following objectives:

1. Determine the structural impacts of invasive black bass on macroinvertebrate communities in three Cederberg mountain streams using standard bio-monitoring metrics and functional feeding group classification.
2. Quantify the functional impacts of invasive bass on the stream food webs by utilizing stable carbon and nitrogen isotope analysis to trace basal resource contributions and estimate the trophic positions of consumers.
3. Evaluate the trophic cascade mechanism by combining taxonomic bio-assessment with isotopic evidence to reveal disrupted energy pathways and altered food web structures.

Together with the systematic review, which maps the current evidence base for stable isotope research in African fresh waters, these findings will help guide invasive fish management, the protection of barrier-isolated refuges, and future research priorities.

2 A natural experiment

Invasive black bass (*Micropterus* spp.) are widespread in the rivers of the Olifants-Doring catchment, where they threaten native, largely endemic fish. However, natural waterfall barriers have prevented their upstream spread in several tributaries, leaving reaches with native fish only immediately above reaches where bass are present.

This study uses these barriers as a natural experiment. On each of three Cederberg rivers (Dwars, Ratel and Eselbank), five sites will be sampled above the barrier and five below, giving 30 sites in total (Figure 2). Environmental variables measured along five transects per site will account for natural upstream-downstream differences in habitat.

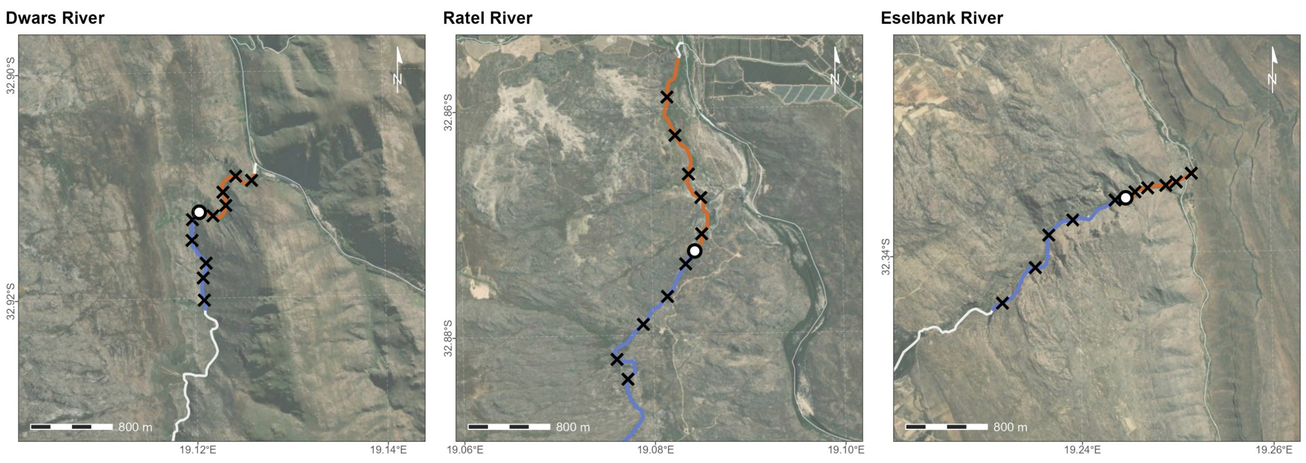


Figure 2. *Approximate sampling sites on the Dwars, Ratel and Eselbank rivers, five above and five below each natural black bass barrier. Locations of bass barriers and reach extents from van der Walt et al., (2016)*
Imagery: Tiles © Esri - Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

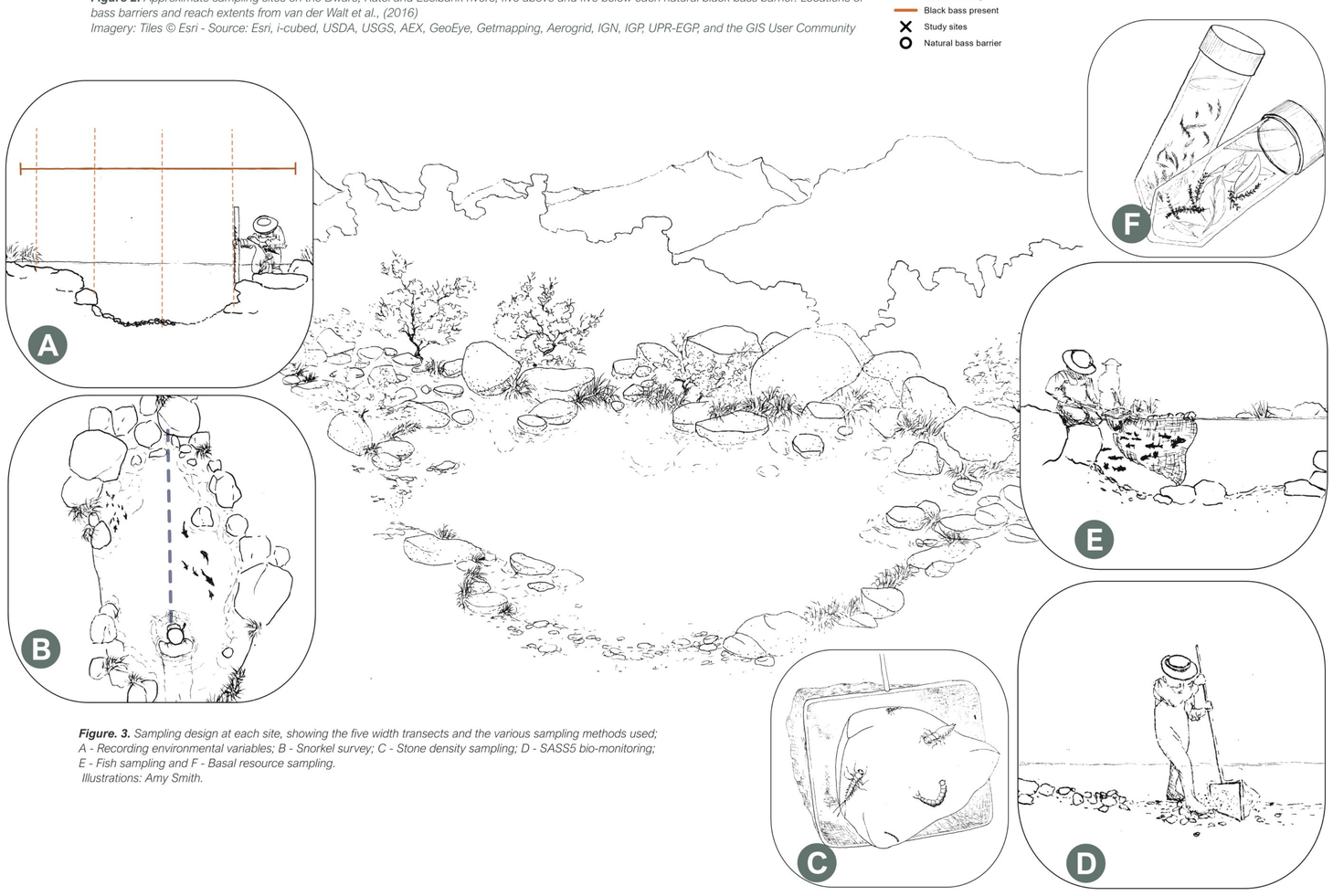


Figure. 3. *Sampling design at each site, showing the five width transects and the various sampling methods used; A - Recording environmental variables; B - Snorkel survey; C - Stone density sampling; D - SASS5 bio-monitoring; E - Fish sampling and F - Basal resource sampling.*
Illustrations: Amy Smith.

3 Methods

In the field:

- **A** Channel width, depth, flow velocity, water temperature, dissolved oxygen, and substrate type recorded along five transects
- **B** Two-pass snorkel fish surveys to estimate relative density and assemblage composition.[3]
- **C** Fine-scale density estimation through stone sampling with a 250 µm mesh net to improve taxonomic and ecological resolution.[7]
- **D** Biomonitoring using the South African Scoring System (SASS) to assess invertebrate assemblage composition.[2]
- **E** Individuals of each fish species will be sampled using a seine net and hand net for gut content analysis (GCA) and stable isotope analysis (SIA).
- **F** Basal resources (biofilm, algae, leaf litter) collected for stable isotope analysis.

In the lab:

- Gut content analysis (GCA) performed on fish to quantify diet.
- Invertebrates identified to genus level and grouped by functional feeding group.
- Fish and invertebrate tissue and basal resource samples dried, ground and analysed for $\delta^{13}C$ and $\delta^{15}N$ by isotope ratio mass spectrometry at UCT Stable Light Isotope Laboratory

Analytical approach:

- Bayesian isotope mixing models to estimate dietary resource contributions.
- Comparative statistical testing of trophic metrics between invaded and uninvaded sites, incorporating habitat covariates.
- Relationships between fish density and trophic responses assessed using regression-based models.

4 Sampling design

3 rivers, 5 + 5 sites each = 30 sites, sampled in summer (November 2026) and autumn (March 2027)

Schematic, not to scale

5 Expected outcomes

This research is expected to generate empirical knowledge on the ecological impacts of invasive black bass on freshwater macroinvertebrate assemblages and food-web structure in mountain river systems of the Cederberg region. Specifically, the study will:

- Provide the first stable-isotope based characterisation of trophic structure and energy pathways in invaded and uninvaded headwater streams of the Western Cape.
- Clarify the mechanisms by which invasive black bass influence macroinvertebrate community composition, functional structure, and ecosystem processes.
- Establish a regional ecological baseline that can be used to detect future environmental change or evaluate restoration success following invasive fish management.
- Contribute applied evidence to support invasive-species control, freshwater biodiversity conservation, and ecosystem management planning in the Cape Floristic Region.

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